

ENGINEERING TRIPOS PART IIA
ELECTRICAL AND INFORMATION ENGINEERING TEACHING LABORATORY
EXPERIMENT 3B2
FPGA PROGRAMING AND TESTING

FULL TECHNICAL REPORT – Guidelines

1. Experiment 3B2 may be written up as a Full Technical Report (FTR).
2. The report must be submitted by **4pm on Wednesday 25 March 2026**. The marked version of the original report must be included. Cover sheets for the FTR are available in the EIETL.
3. General information on writing Full Technical Reports is provided in the 'Part IIA Coursework & Labs overview'. This is accessible on the web, at:
<http://teaching.eng.cam.ac.uk/content/part-ii-coursework-labs-overview>
4. You should make sure that you have done the optional parts of the experiment. You may need more time in which to collect this extra data. The difference between the normal lab write up and this report is in the thoroughness of the extraction of results, the interpretation of the data and the depth of the conclusions drawn (particularly including the 'extra' data).
5. In particular, you should consider the following:
 - a) Comment on the features of the FPGA board you choose to use. Describe the clock options available on the board and how you made use of.
 - b) Describe the assignment of the circuit inputs and outputs to specific pins on the FPGA, and comment on its configurability.
 - c) Comment on the design of the circuit you choose to configure and the behaviour of the FPGA board you observed.
 - d) Comment on the resources (number of logic elements, registers and pins) needed for your designed circuit.
 - e) Whenever possible, apply theory to explain the behaviour of the logic circuits you used during the experiment.